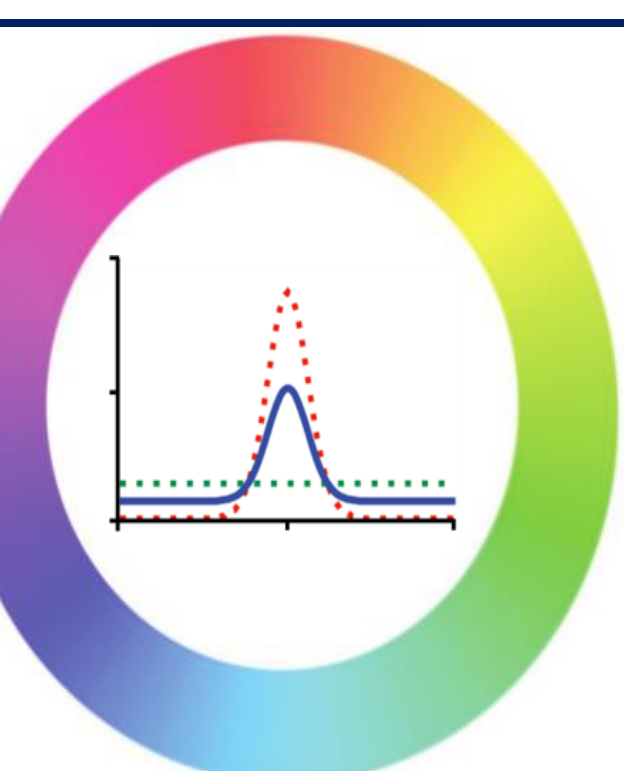




Direct Competition between Working Memory Precision and Hippocampal Pattern Separation



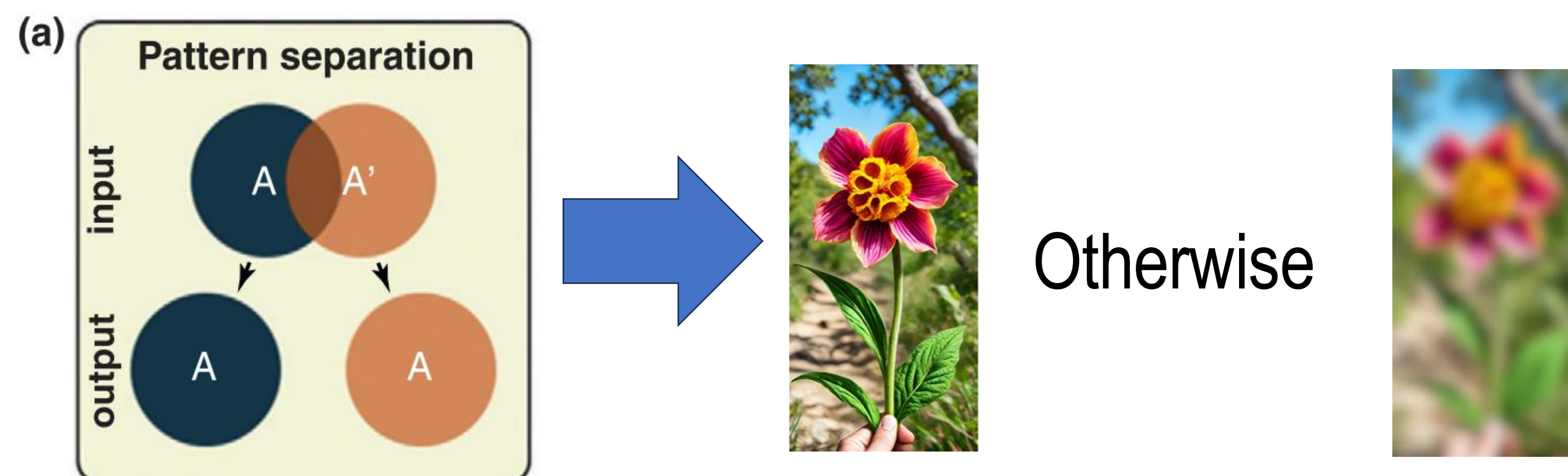
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INTRODUCTION

Overall Aim

- Testing pattern separation hypo. for working memory precision
 - WM precision is supported by hippocampal pattern separation.



Working Memory Precision

- Correspondence between external stimuli & internal representations

Hippocampal Pattern Separation

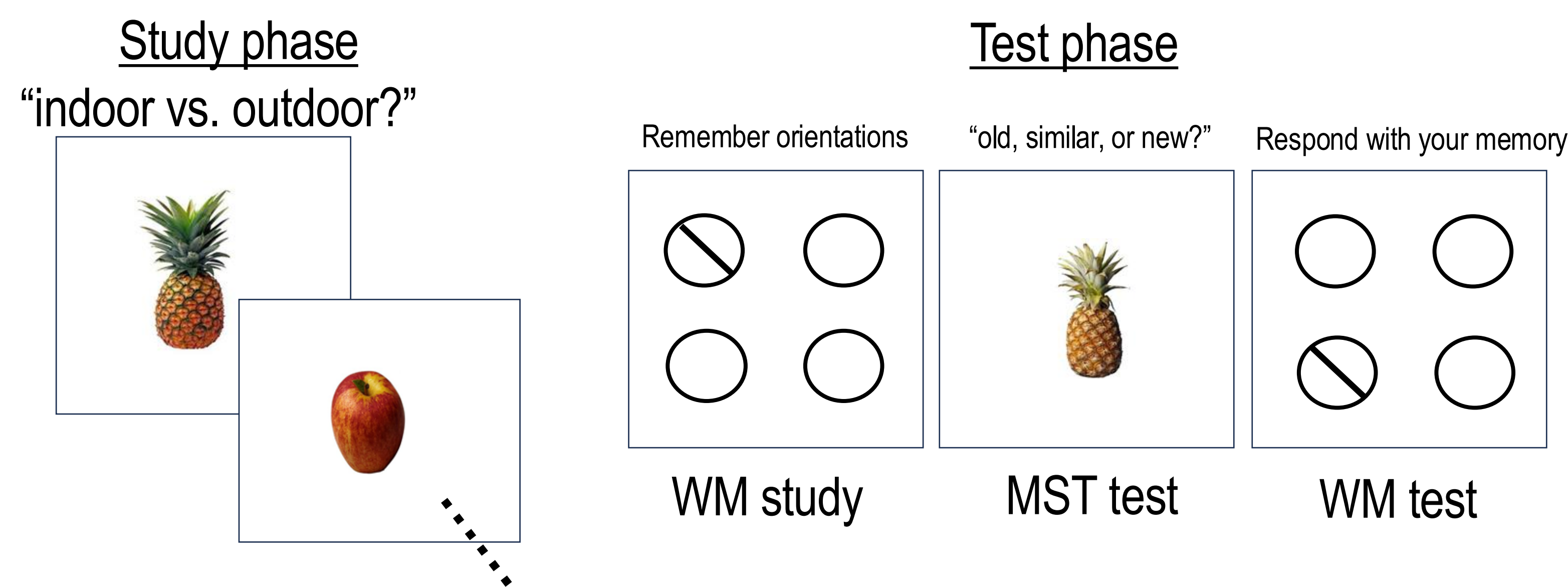
- Neurocomputational process of separating similar inputs into non-overlapping ones in the hippocampus¹

Supporting Evidence

- WM representations of simple features decoded from hippocampal activity²
- Damage or stimulation on MTL impaired WM precision³

The present study

- Testing the direct competition between pattern separation and working memory precision



MST trial types

- Target** – Same as the studied item
- Lure** – Similar, but not the same (high pattern separation load)
- Foil** – Not seen at study

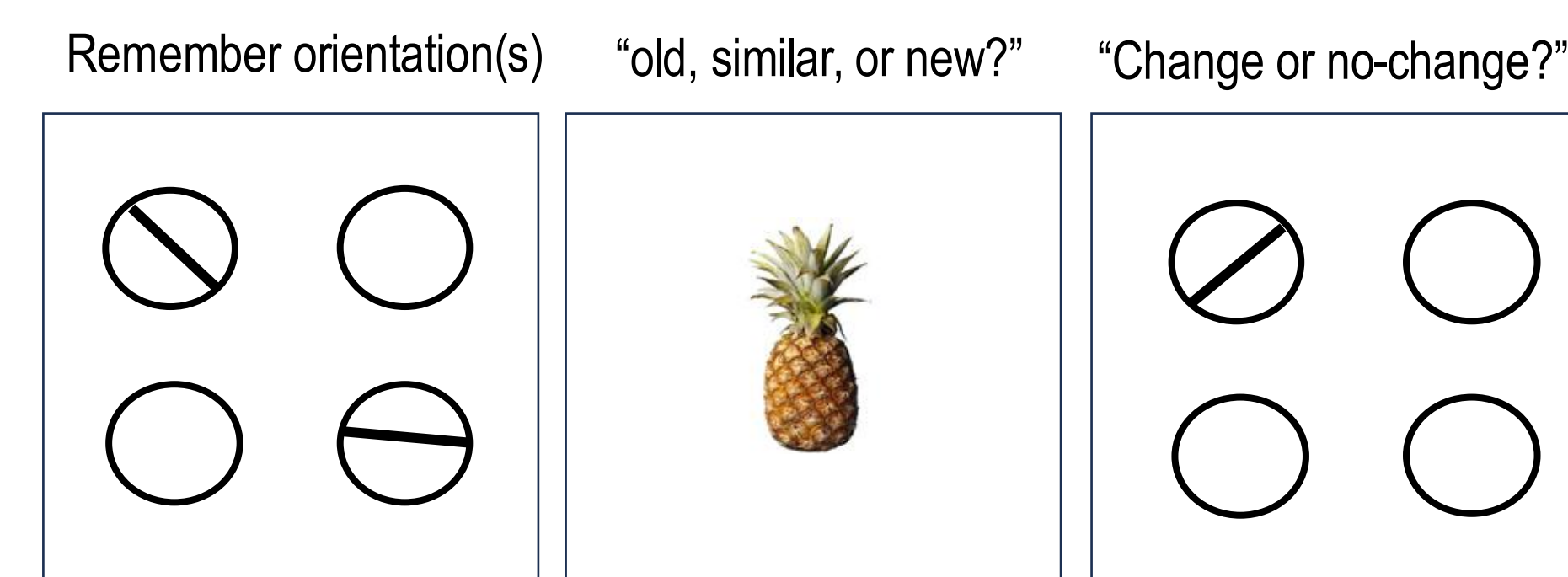
The Mixture model (slot model)^{4, 5}

- WM retrieval are a mixture of likelihood of successful memory (capacity) and noisy representation (precision)

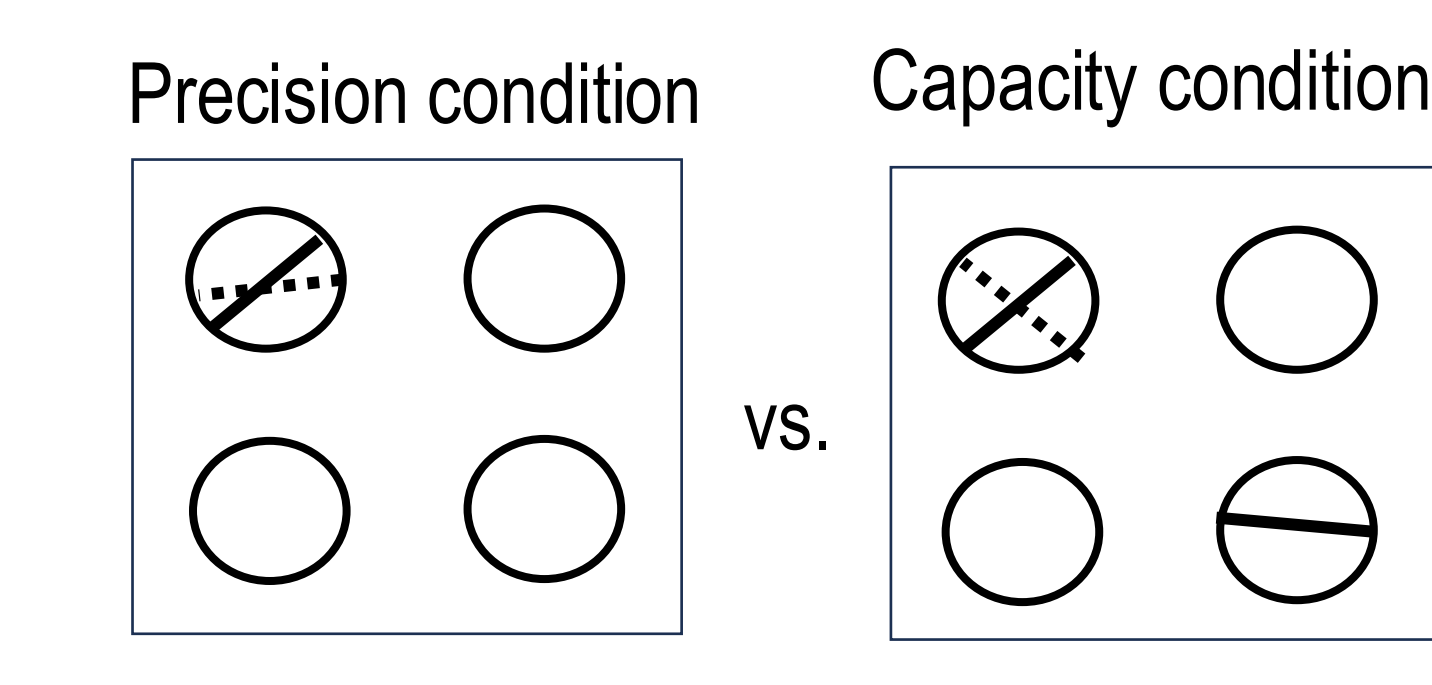
METHODS AND RESULTS

✓ Exp1: How does pattern separation demand affect WM performance

Change Detection task



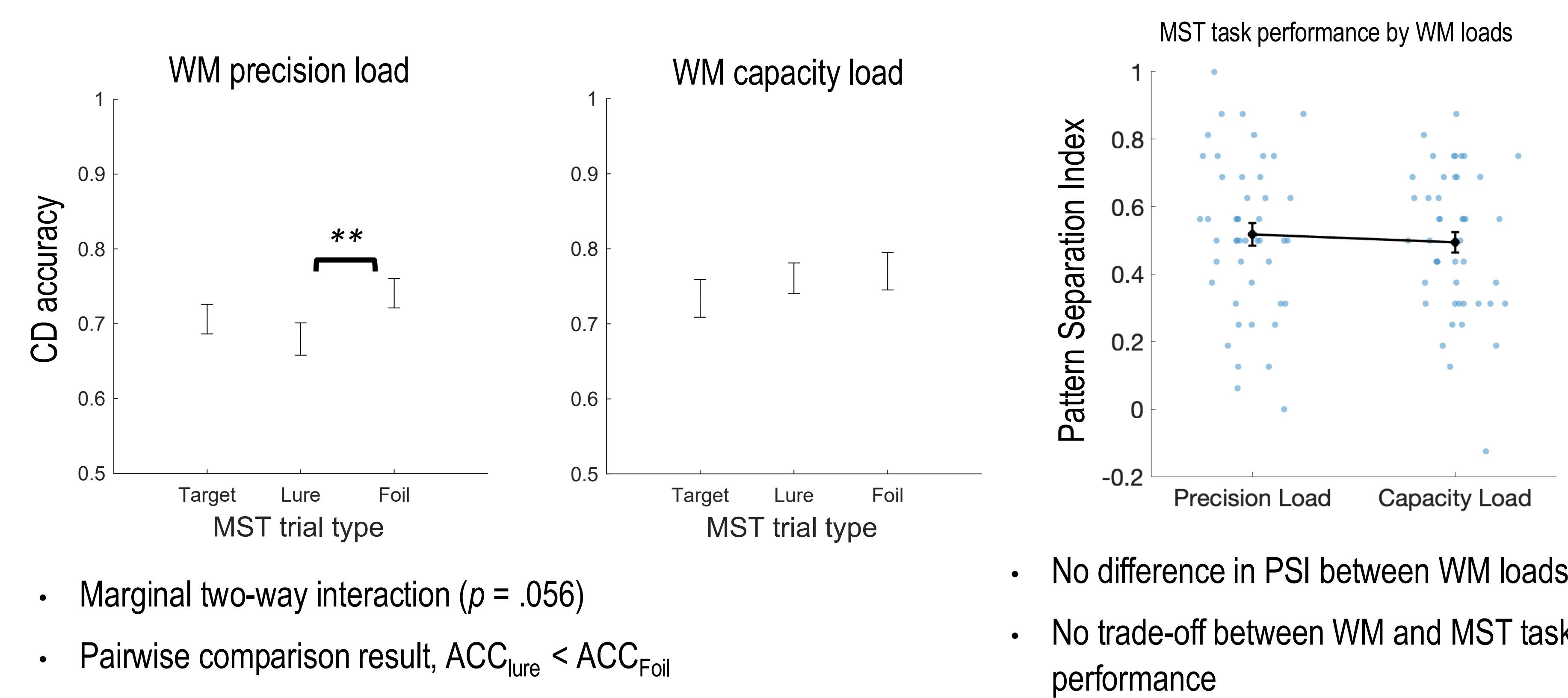
Manipulations



Set size 1 (small change) vs Set size 2 (large change)⁶

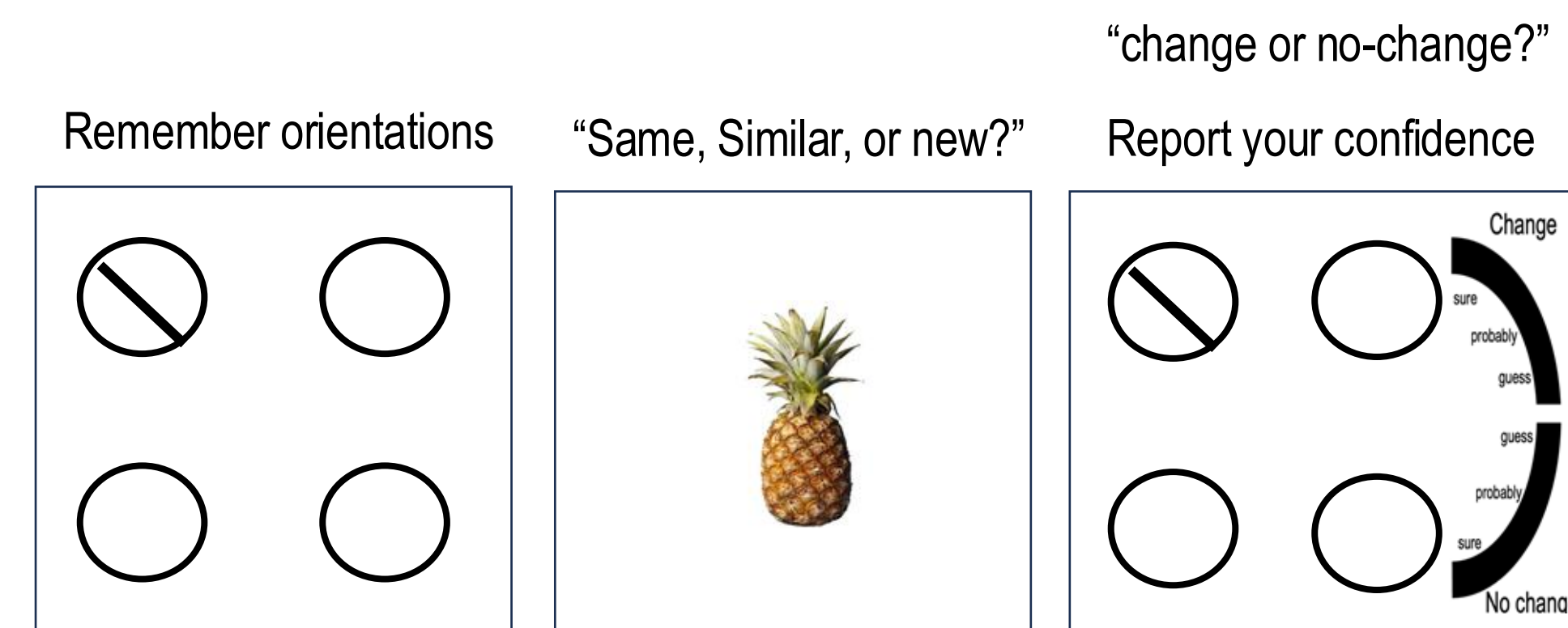
Prediction: CD performance will be reduced during MST lure trials, under WM precision condition

Results

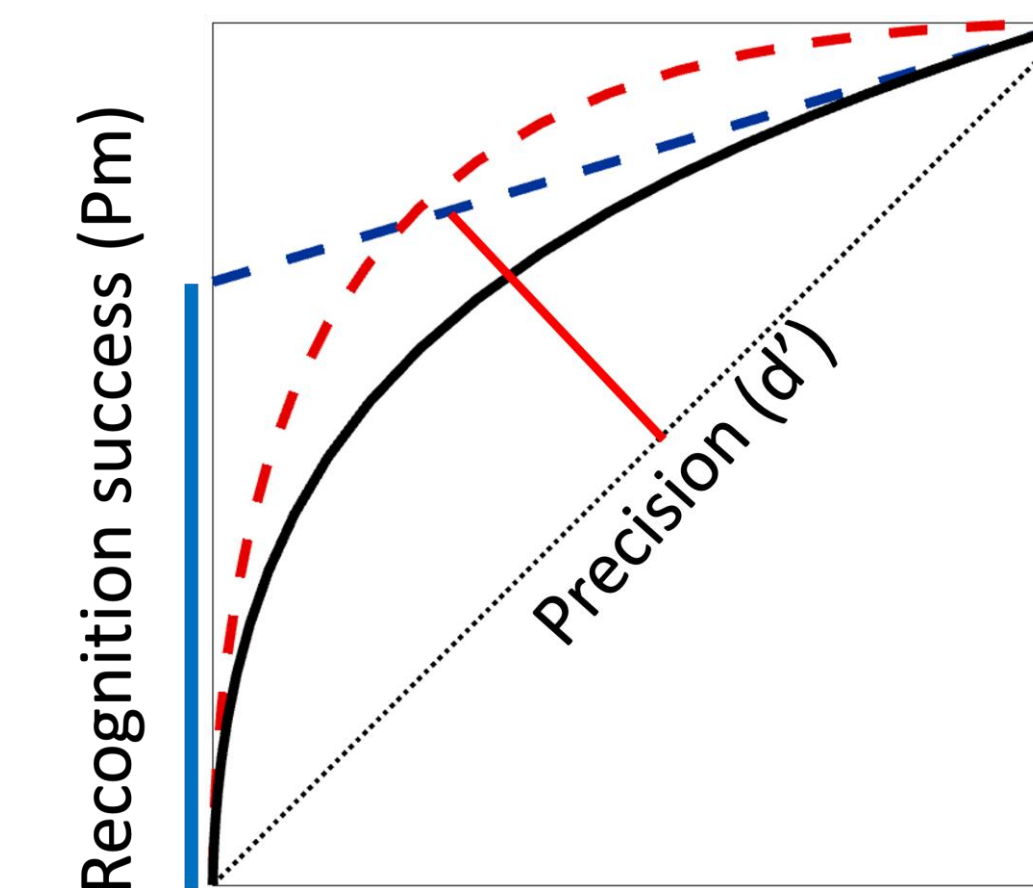


✓ Exp2 : Is direct measure of WM precision reduced during MST lures?

Receiver Operating Characteristics

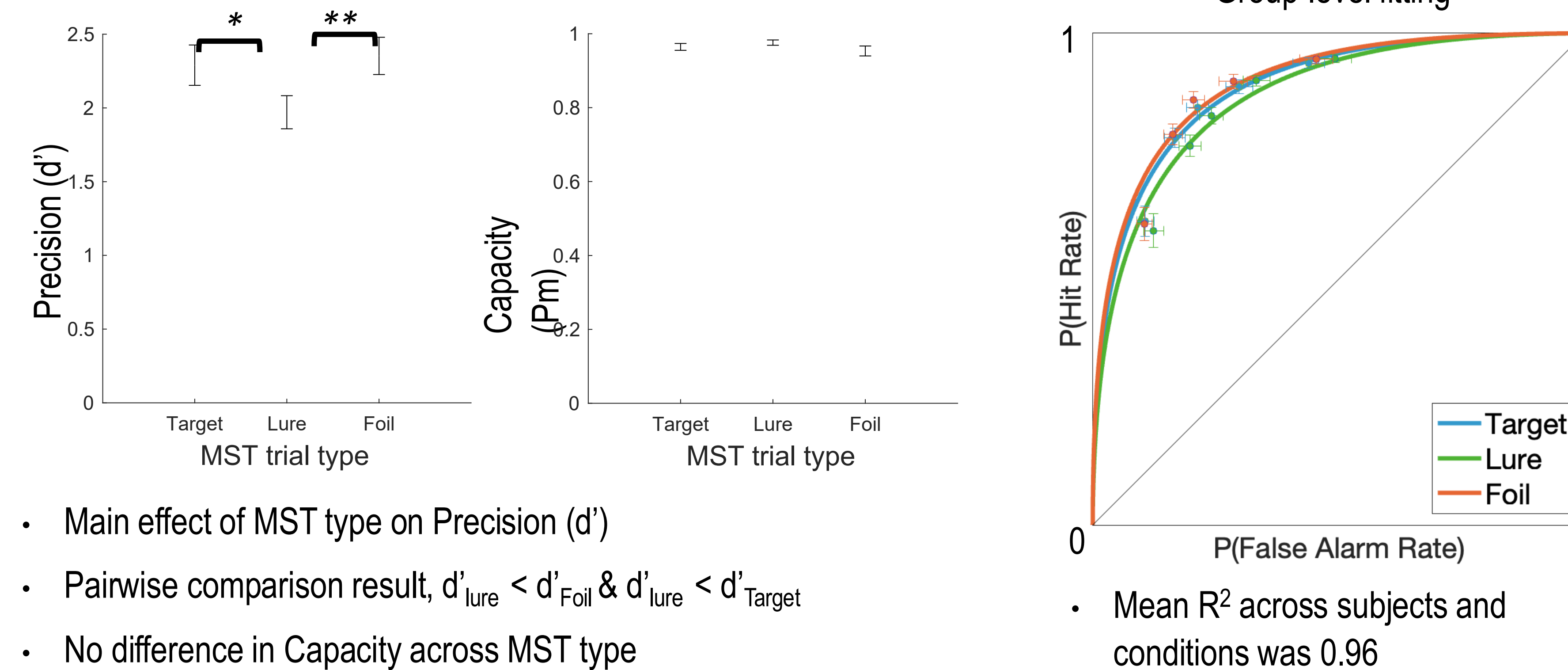


ROC Mixture Model⁵



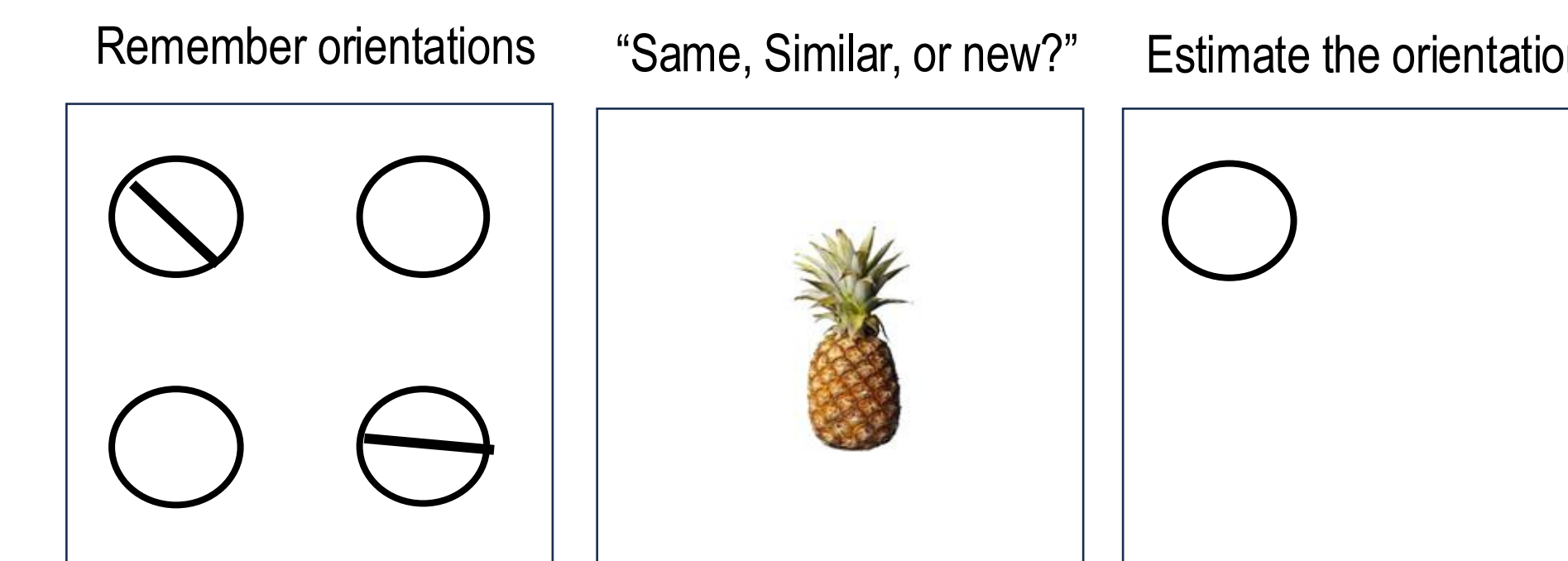
Prediction: WM precision will be reduced during MST lures, whereas capacity will be intact

Results

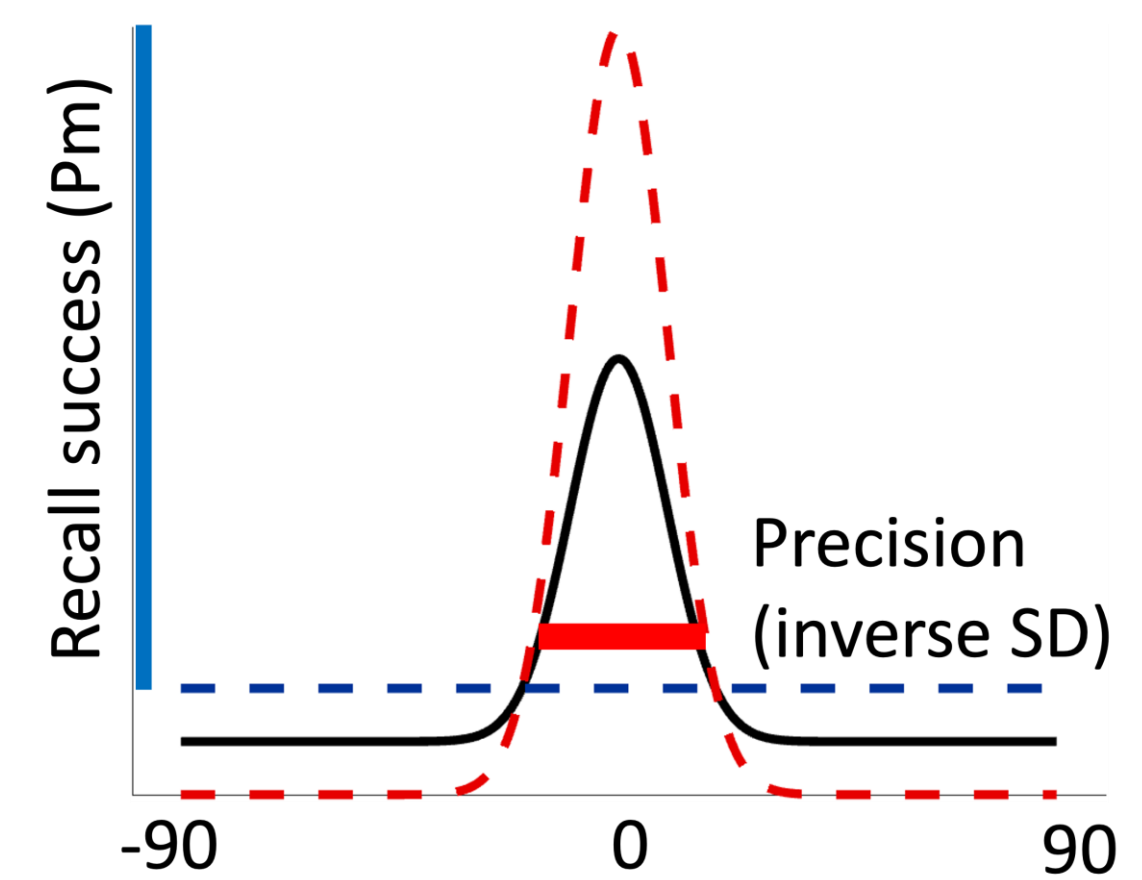


✓ Exp3 : WM precision is also reduced during MST lures using recall paradigm?

Recall task

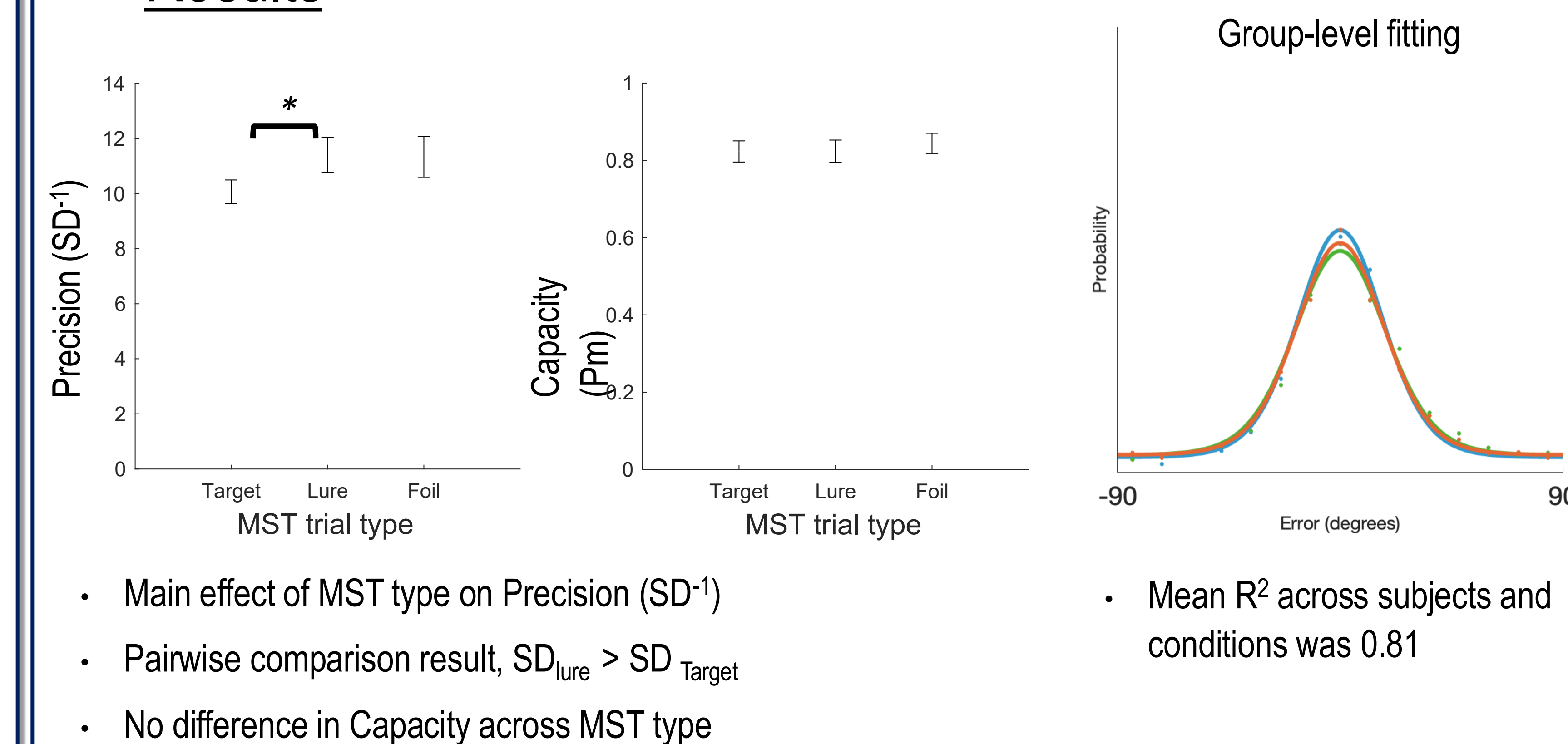


Recall Mixture Model⁴



Prediction: WM precision will be reduced during MST lures

Results



✓ Discussion and Conclusions

- Reduced WM precision under high pattern separation load, but intact WM capacity, across three experiments
- Converging evidence from various methods supporting *hippocampal pattern separation hypothesis* of WM precision.

✓ Future studies

- Does perceptual discrimination process compete with WM precision?

✓ References

- Yassa, M. A., & Stark, C. E. L. (2011). Pattern separation in the hippocampus. In Trends in Neurosciences, 34(10), 515–525.
- Xie, W., Cappiello, M., Yassa, M. A., Ester, E., Zaghloul, K. A., & Zhang, W. (2023). The entorhinal-DG/CA3 pathway in the medial temporal lobe retains visual working memory of a simple surface feature. Elife, 12, e83365.
- Xie, W., Chapeton, J. I., Bhasin, S., Zawora, C., Wittig Jr, J. H., Inati, S. K., ... & Zaghloul, K. A. (2023). The medial temporal lobe supports the quality of visual short-term memory representation. Nature Human Behaviour, 7(4), 627–641.
- Zhang, W., & Luck, S. J. (2008). Discrete fixed-resolution representations in visual working memory. Nature, 453(7192), 233–235.
- Xie, W., & Zhang, W. (2017). Dissociations of the number and precision of visual short-term memory representations in change detection. Memory and Cognition, 45(8), 1423–1437.
- Zhang, W., & Luck, S. J. (2015). Opposite effects of capacity load and resolution load on distractor processing. Journal of Experimental Psychology: Human Perception and Performance, 41(1), 22.